Jacky Song

Computational physicist & developer

Founder · Research · Computational Physics · Designer · Web & Software Engineer · Artist · Musician

Education

2023 - 2026 **B.S. in Applied Physics, Rensselaer Polytechnic Institute**

Graduated *cum laude* with a degree in Applied Physics from Rensselaer, with a concentration in computational quantum physics.

Research

2021 - Present **Ultra-long-distance power transmission and space-based energy capture**

As lead researcher of the Project Elara team, I conducted research into modeling electromagnetic power transmission for space-based solar power systems and designing high-power free-electron lasers using Monte-Carlo simulations.

2025 **Machine learning-assisted material prediction and simulation for laser active media**

I utilized machine learning on the AiMOS supercomputer to be able to predict materials optimally-suited for laser gain media for building next-generation lasers. My research was conducted as part of the Materials Intelligence research group with collaborators at UCLA.

Work experience

Nonprofit head	I founded and am the Head of **Project Elara**, a nonprofit open-source organization dedicated to advancing space-based solar power research.
Research lead	As part of Project Elara, I lead a multidisciplinary research group conducting physical testing and computational modelling of lasers, spacecraft, and wireless power transmitters. I created several libraries for high-performance computing and graphics.
RCOS coordinator	I am a coordinator and student leader at the Rensselaer Center for Open-Source (RCOS) and acted as a Teaching Assistant for the CSCI-1700, CSCI-2700, CSCI-4700 and CSCI-4710 courses.
Physics mentor & educator	I assisted faculty, helped out with experiments, and mentored students in the PHYS-1250 course at RPI. I have also written several open-access textbooks on physics.
Visual & web designer	I have worked on multiple website designs for several open-source projects, including Natron and Mewa, and made custom designs for the RPI Physics Department.

Skills

Specializations	Computational physics, electromagnetics, laser physics, mathematical modelling, numerical simulations, open-source development, scientific education & outreach
Programming languages	C, Python, Rust, GLSL, MATLAB/Octave, Mathematica, HTML, CSS, Sass, JavaScript, Typescript, LaTeX, Typst
Technologies	NumPy/SciPy/Pandas, Jupyter, CMake, OpenGL, Jekyll, Zola, Flask, NodeJS/NPM, Tailwind, Linux, Bash, SSH, Git, Numba, TensorFlow/Keras, Scikit-learn, Optuna, Streamlit
Technical skills	Finite difference & finite element simulations, high-performance computing, graphics rendering, machine learning, full-stack web development
Software	MATLAB, Mathematica, FreeFEM, Figma, Blender, Inkscape, DaVinci Resolve, OpenSCAD
Creative skills	Video editing, 3D modelling & graphics, 2D vector graphics, VFX compositing, instrumental performance, music composition, UI/UX design, logo design, graphic design